

PHY201 Practice Exam on chapter(s): 4, 5, 6

Name: _____

These questions assess your understanding of the material from Chapters 4, 5, 6 of OpenStax College Physics. Please write your name on the sheets of paper that you use and show all your work.

1. Michael places a 21.1-kg orange wooden block on a wooden table. He then attaches a pulley to the table, threading a rope through the pulley to a 8.6-kg red block. This red block is hanging in the air by the rope and does not make contact with anything. What is the acceleration of the red hanging block? Note that for wood on wood, $\mu_s = 0.5$ and $\mu_k = 0.3$.
ANSWER: _____
2. Allison, who has a mass of 69.0 kg, steps onto a scale that rests on the floor of an elevator. If the elevator accelerates downward at a rate of 2.4 m/s^2 , what will the scale read?
ANSWER: _____
3. William is decelerating as he snowboards at 3.5 m/s up a slope that is inclined at 25.7° above the horizontal. Calculate the magnitude of his deceleration. The total mass of William and his snowboard is 76 kg. The density of the air is 1.277 kg/m^3 . Note that for waxed wood on snow, $\mu_s = 0.14$ and $\mu_k = 0.1$. Ignore the effect of air resistance.
ANSWER: _____
4. When rebuilding his car's engine, Sam finds that he must exert 270 N of force to insert a dry steel piston into a steel cylinder. What is the magnitude of the normal force between the piston and cylinder as the piston is inserted into the cylinder at a constant speed? Note that for dry steel on dry steel, $\mu_s = 0.6$ and $\mu_k = 0.3$.
ANSWER: _____
5. Jennifer is a contestant in a winter sporting event. She is tasked with pulling a rope connected to a 330-kg block of ice on a frozen lake. Jennifer pulls diagonally upward on the rope at an angle of 21° from the vertical. What is the minimum force at which she must pull in order to get the block of ice moving? Note that for ice on ice, $\mu_s = 0.1$ and $\mu_k = 0.03$.
ANSWER: _____
6. Chloe goes outside on a summer morning with her tea and sees a perfectly vertical strand of spider web. She watches for a while and eventually a spider climbs up the strand. At this point, invested in her experiment, Chloe takes the spider and puts it on a scale, measuring its mass to be 343 mg, then returns it to its strand of web. What is the tension in the web when the spider is sitting on it?
ANSWER: _____
7. While pitching in a baseball game, Julia brings her arm forward during a pitch, rotating her forearm about her elbow. If the velocity of the ball in her hand is 29.6 m/s and the ball is 0.26 m from the elbow joint, what is the angular velocity of the pitcher's forearm?
ANSWER: _____
8. Two nurses, Allison and Alexis, work together to push a crash cart by pushing on the handle. They push it from rest so that it covers a distance of 9.9 m in 5.1 s. Allison exerts 200 N at 39.3° below the horizontal and Alexis exerts 170 N at 24.5° below the horizontal. The force of friction is 214 N. Determine the mass of the cart.
ANSWER: _____
9. Natalie is riding a motorcycle that can produce an acceleration of 4.4 m/s^2 while travelling at 90 km/h. The magnitude of the two forces resisting motion, friction and air resistance, total 700 N. The mass of Natalie and the motorcycle is 270 kg. What is the magnitude of the force that the motorcycle's rear tire exerts on the ground?
ANSWER: _____
10. Given that the planet Aurora orbits the planet Xanadu once every 59.2 days and that it is an average distance of 197,000 km from the center of Xanadu, calculate the period of an artificial satellite orbiting Xanadu at an average altitude of 2,300 km above Xanadu's surface. Note that the radius of Xanadu is 14,500 km.
ANSWER: _____
11. A team of eight huskies pulls William and his sled across snow at 4.8 m/s . William and his loaded sled have a combined mass of 290 kg and the average mass of each dog is 24.0 kg. If each dog can pull with a force of 150 N, what is the magnitude of the acceleration of the sled? Note that for waxed wood on snow, $\mu_s = 0.14$ and $\mu_k = 0.1$.
ANSWER: _____

12. Ryan pushes a crate made of vorpalium on a floor made of mythralium. The crate is initially at rest on the horizontal floor, until Ryan pushes horizontally on the crate with a force of 260 N, the minimum force necessary to start its motion. Find the mass of the crate. Note that for vorpalium on mythralium, $\mu_s = 0.43$ and $\mu_k = 0.63$.
ANSWER: _____
13. Olivia is trying to keep a 39.4-kg wagon at rest on a frictionless inclined plane. The plane makes an angle of 79.6° with the horizontal. What is the magnitude of the force Olivia must apply on the wagon to keep it from sliding down the plane?
ANSWER: _____
14. Brandon is not only a part-time guide at the Battleship USS Iowa Museum, but also a physics student. He wants to find the net external force that would be exerted on a 950-kg artillery shell fired from one of the USS Iowa's turrets. Brandon measures the barrel of a turret to be 16.8-m long and reads that a shell, fired from rest, would leave the barrel at 750 m/s. What is the net external force exerted on the shell?
ANSWER: _____
15. A 964-kg rocket ship, travelling at 90 m/s, is acted upon by an average force of 29 kN applied in the direction of its motion for 21 s. What is the change in the rocket's velocity?
ANSWER: _____
16. Lauren drives her 1490-kg car around a 370-m radius unbanked curve at 27.5 m/s (about 61 mph). Lauren's mass is 80 kg. What is the magnitude of the centripetal acceleration of Lauren's car? For reference, accelerations over approximately 0.3 m/s^2 are mildly uncomfortable.
ANSWER: _____
17. A 20-kg child is riding a playground merry-go-round that is rotating at 7.5 revolutions per minute. What centripetal force must she exert to stay on if she is 8.0 m from the center of the merry-go-round?
ANSWER: _____
18. A flea jumps into the air by exerting a force straight down on the ground. A breeze blows on the flying flea and exerts a horizontal force of $1 \mu\text{N}$ on the flea. Find the magnitude of the acceleration of the flea if its mass is 2.94 mg.
ANSWER: _____
19. Shiv observes a 571-g squirrel with a cross-sectional area of 292 cm^2 falling from a large tree to the ground. The temperature of the air is 25.2°C . Recall that the density of air ρ (in kg/m^3) depends on temperature T (in $^\circ\text{C}$) according to $\rho = \frac{353}{T+273}$. Assume the squirrel's drag coefficient of 1.0. Assuming the squirrel reaches terminal velocity, at what speed will it hit the ground?
ANSWER: _____
20. Jessica is an astronaut walking on the surface of the planet Zephyria. She experiences a small acceleration due to Earth's gravitational field, even at the distance between the center of Earth and Jessica of 0.39 Gm. What is her acceleration due to Earth's gravitational field?
ANSWER: _____